

Subhadeep Sarkar

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SUMMARY

Computer Science and Engineering undergraduate specializing in AI and Machine Learning. Proficient in Java, Python, and C++ with strong expertise in data structures, algorithms, and software development. Adaptable team player recognized for analytical problem-solving and clear communication of technical concepts.

TECHNICAL SKILLS

Programming: Java(4/5), Python(4/5), C++(3/5)

Web Development: HTML(4/5), CSS(3/5), basics of JS(2/5)

Cloud: Amazon Web Services (AWS)

Concepts: Artificial Intelligence, Machine Learning, Deep Learning, Data Structures, Algorithms

Other: Problem Solving, Analytical Thinking, Debugging, Collaboration, Communication

Frameworks: Django, Pandas, Pytorch, Matplotlib

PROJECTS

Biomedical Knowledge Graph from Literature

2024

Python, BERT (Sequence Classification, graph mining)

- Developed a knowledge graph pipeline that extracted and classified over 10,000 unique biomedical relationships from scientific literature, improving discovery of COVID-19-related entity interactions by 30% compared to manual methods. `pyecharts`.
- Automated data preprocessing and relation extraction workflows, reducing manual analysis time by 70% and enabling real-time visualization for faster hypothesis generation among researchers.

Forest Fire Detection using Machine Learning and IoT

2024

YOLOv3, CNN, Python, Raspberry Pi scripting

- Engineered a real-time forest fire detection system that increased early detection accuracy by 25% using deep learning on drone imagery and IoT sensor data, compared to traditional manual monitoring.
- Reduced alert response time from detection to notification to under 30 seconds (previously 5 minutes), via edge processing and cloud-based dashboards.
- Enabled scalable remote monitoring over a 10 sq km test site, supporting rapid emergency response and minimizing fire damage.

EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science and Engineering (AI/ML)

- GPA: 8.76/10

Aug 2023 – May 2027 (expected)

Bhopal, India

Modern Higher Secondary School

Higher Secondary (Science)

Jun 2021 – Jan 2023

Agartala, Tripura, India

Holy Cross School

Primary and Secondary Schooling

Jan 2010 – Jan 2021

Agartala, Tripura, India

CERTIFICATIONS

Explore AIML with Crowdsourcing

Fundamentals of AI and ML

Python Essentials

AWS Cloud Clubs Machine Learning Camper

Python Basic Programs